

University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering
Problem Set 4.2: Rough Root Locus

Question 1

For systems with the poles and zeros described,

- i) Sketch a rough root locus for the system (as in, just the general shape)
 - ii) Is it possible for the closed-loop system to become unstable?
 - iii) Is it possible for the closed-loop response to become oscillatory?
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- a) Poles at $-2 \pm 3j$
 - b) Poles at -5 and -6 , zeros at -2 and 1
 - c) Pole at -4 and zero at 2
 - d) Poles at -3 and -1 , zeros at $1 \pm j$
 - e) Poles at $-1 \pm 5j$, zero at -5